

Forward Curve Modelling in Commodity Markets

Svetlana Borovkova and Hélyette Geman

2.1 INTRODUCTION

Commodity markets have been buoyant for the last few years and commodity spot prices continue (at the time of writing) to reach new record highs, driven by a strong world demand and a tight supply, very sensitive to political or weather events. Moreover, for more than a century, forward markets have been the main place of commodity trading as they allow investors to avoid the hurdles of physical delivery while getting the desired exposure to changes in commodity prices. Most examples in this chapter dedicated to the fundamental problem of forward curve modelling will be investigated in the setting of strategic energy commodities such as crude oil or natural gas. But the methodology identically applies to other commodity classes, either seasonal such as agriculturals or non-seasonal such as metals.

First we recall some elementary definitions around forward and futures contracts. *Forward contracts* are traded over-the-counter between two parties and the grade of oil, delivery point, exact amount and exact delivery date are specified at the time of writing the contract. In this respect they are different from *futures contracts*, which are standardized in terms of the grade, amount, delivery date and location. Energy futures are traded on exchanges such as the InterContinental Exchange (ICE) and New York Mercantile Exchange (NYMEX). For instance, most of the trading activity in oil markets takes place in futures and forward contracts, where the volume of trades is almost ten times higher than in the spot market.

Throughout the chapter, we will denote the futures or forward price on day t for maturity date T by $F(t, T)$ and the spot price on day t by $S(t)$. The forward curve prevailing on day t is the collection of futures prices $\{F(t, T)\}_T$ for all traded maturities $T = 1, 2, \dots, N$.

In what follows, we will use indifferently the terminology futures or forwards. It has been known for some time (see Cox *et al.* (1985)) that, in the absence of credit risk, forward and futures prices are equal under non-stochastic interest rates. In commodity markets, interest rate risk plays a secondary role compared to the commodity spot price risk and will not be our first concern in this chapter (see Geman and Vasicek (2001) for the introduction of

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stochastic interest rates in energy forward and futures contracts). From now on we will use terms “futures price” and “forward price” interchangeably.

For crude oil, futures for up to 84 months are traded on both the ICE and NYMEX, for oil products such as heating oil and gasoline up to 18 and 36 months respectively. The NYMEX recently started trading long-dated natural gas futures contracts, going up to 72 months ahead. More and more long-dated energy futures contracts are being traded today, resulting in the availability of longer forward curves.

The first-to-expire futures contract is usually referred to as the *front-month*, or *nearby* contract, and denoted by the letter M after the contract's name. The second nearby contract (next-to-front) is denoted by M+1, the next one M+2 and so on. On the first maturity date, the front-month contract expires and the next-to-front becomes the front-month contract. All other futures contracts shift by one position (M+2 becomes M+1 and so on) and a new contract is added. This shift is called the *rollover*. It plays a key role in the management of commodity indices as the roll yield may be a positive or negative component in the total return, depending on the shape of the forward curve. Given the billions of dollars that have been recently poured into commodity markets, the rolling activity has itself an impact on the volatility of the nearby futures. This in turn may induce a portfolio manager to invest the share of the index dedicated to a given commodity (e.g. crude oil) into more distant maturity forward contracts.

Futures markets such as the ICE and NYMEX provide the most reliable and liquid forward curves. However, sometimes it can be beneficial to use OTC forward prices for the construction of the forward curve. For example, in electricity markets this can be a better alternative, as futures contracts are either not traded on exchanges or are rather illiquid. Then it can be a complicated task to construct a smooth forward curve, indexed by regular maturities, from prices of irregularly spaced and complex forward contracts (which may be based for instance on average prices over a specified period). The example of such a forward curve construction is given in Benth *et al.* (2007).

Forward curves are of paramount importance in commodity markets, for several reasons. They provide information about the views of market participants, anticipated price trends and expectations about future supply and demand. Futures prices observed in liquid futures markets provide price discovery and are essential for daily marking to market existing portfolio positions as well as for risk management activities such as VaR calculations. Forward commodity prices influence storage, production and other strategic decisions in related industries such as oil refining or soybean processing. Finally, futures contracts provide the right way of calibrating derivatives pricing models under the risk-neutral probability measure, as they present a great liquidity, hence the market “view” one needs to capture.

The “Rational Expectation Hypothesis”, discussed for a long time by famous economists (such as Keynes and Lucas), examines whether forward prices provide valid forecasts for spot prices in the future. However, the actual forecasting ability of futures prices in commodity and, in particular, in oil markets is rather poor. For example, if futures prices were right in predicting the spot price of oil then, according to the forward curves observed one and two years ago (February 2006 and 2007), the oil price in February 2008 should have been 67 \$/bbl (while in fact, on 19 February 2008, it was over 100 \$/bbl).

The spot price of a commodity is often considered as the most important factor driving the whole forward curve (see Gibson and Schwartz (1990), Schwartz (1997)). The influence of the spot price becomes greater as the futures contract approaches maturity, culminating in the following convergence property: on maturity date, the futures price must coincide

with the spot price

$$F(T, T) = S(T). \tag{2.1}$$

This property follows from the absence of arbitrage and the possible occurrence of physical delivery of a commodity at the futures contract’s maturity. However, there are markets where the spot price is rather opaque or unreliable (or completely non-existent, as is the case, for instance, in electricity markets). In such cases, the convergence property does not necessarily hold. It is also possible that the convergence does not take place smoothly as the maturity approaches (which is the case in, for instance, crude oil markets), but only on maturity date, when the futures price makes a sudden move to coincide with the spot price. These considerations, among others, may question the suitability of the spot price as the main driving factor of the forward curve. We shall return to this issue in Section 2.3.

Crude oil forward curves have traditionally been in one of these two shapes: backwardation, when the futures prices for short maturities are more expensive than those maturing later, or contango, which is the opposite situation. Figure 2.1 shows the NYMEX crude oil forward curve on 21 February 2008, when the market was in backwardation, and Figure 2.2 shows a contango forward curve observed on 28 February 2007.

Whether the market is in backwardation or in contango depends on the current price as well as inventory levels, transportation and storage costs, supply and demand equilibria, strategic and political issues and many other factors. A particular shape of the oil forward curve (backwardation or contango) is closely related to the notion of the so-called *convenience yield*, i.e. the benefit of holding a physical commodity over a futures contract (in the

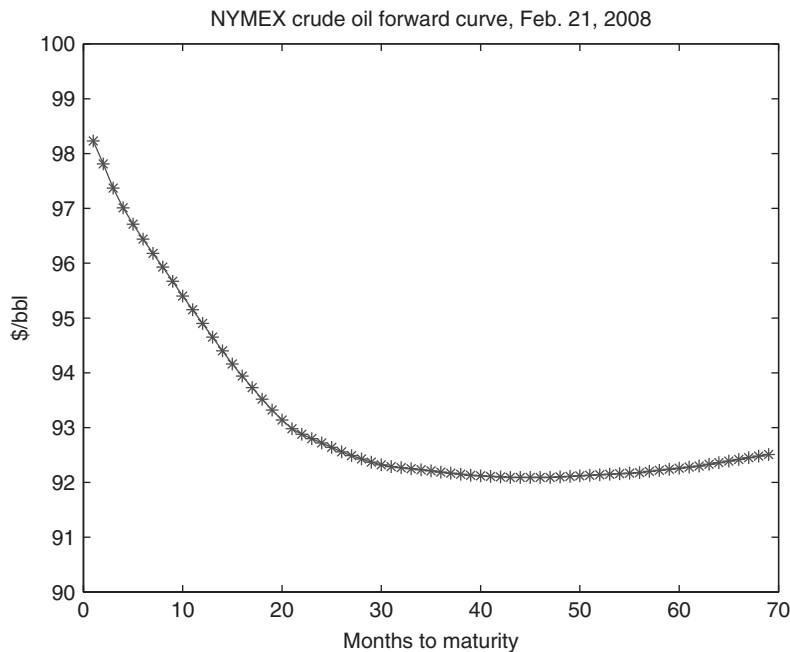


Figure 2.1 NYMEX crude oil forward curve, 21 February 2008

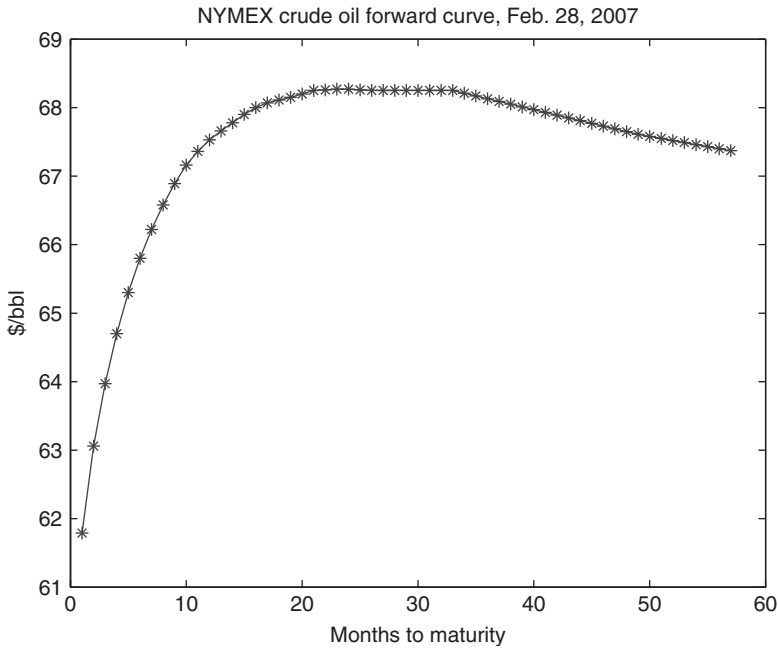


Figure 2.2 NYMEX crude oil forward curve, 28 February 2007

next section we will discuss the convenience yield in more detail). Backwardated forward curves arise when the benefit of holding a physical commodity (i.e. the convenience yield) is high and the interest rates and storage costs (representing together the “cost of carry”) are relatively low. This happens, for instance, when there is a general perception of low availability of oil and instability of its supply. More generally, the shape of the forward curve summarizes perceptions of oil market participants about the current state and future developments of the global oil market, such as new fields discovery or new deep sea oil rig equipment.

As said before, oil forward curves have until recently always been in either backwardation or contango state, with a much more frequent situation of backwardation (see Gabillon (1995)), in agreement with the terminology “normal backwardation”, introduced in the economic literature by Keynes. However, in 2005–2006 a hump-shaped forward curve was observed briefly (see Figure 2.3), which was probably caused by a massive arrival of hedge funds and other new investors into oil futures markets and the short-dated futures rollover strategies they employed.

Lastly, for seasonal commodities such as natural gas, electricity, or agricultural commodities, futures prices are largely governed by seasonal demand (as is the case for energy) or supply (for agricultural commodities). For those commodities, transportation or storage capabilities are limited, so the excess demand or supply shortage cannot be absorbed by transporting a commodity from a different part of the world with excess supply, or storing at times of plentiful supply and using it whenever necessary. This results in a high seasonal premium on futures contracts maturing during periods of high demand (such as winter, in

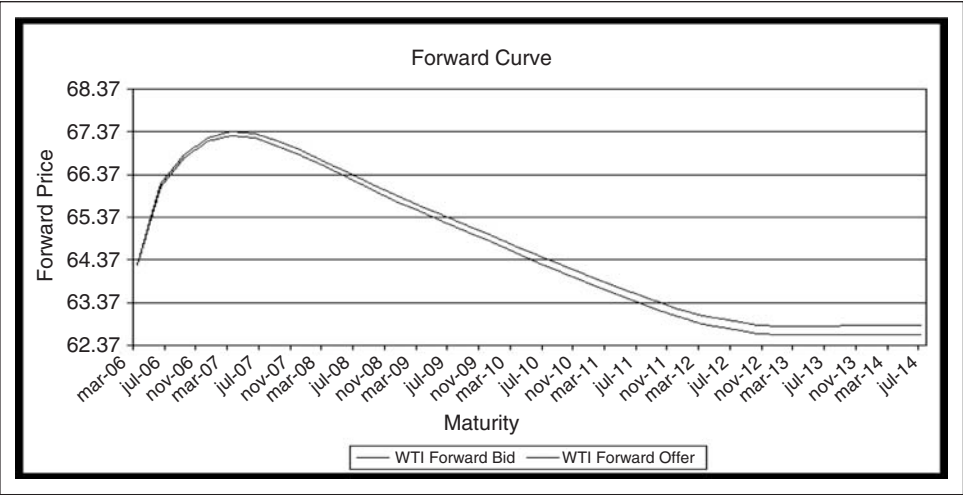


Figure 2.3 NYMEX crude oil forward curve, 7 March 2007

the case of natural gas) or low supply (such as before harvest, for agricultural commodities). This is illustrated in Figure 2.4, where a forward curve for natural gas in the UK is shown. Futures maturing in the winter (the time of high demand for gas in the UK) are clearly at a premium compared to those maturing in summer.

Most traditional methods of forward curve modelling, which we will review in Section 2.2, are unable to deal with this prominent seasonal feature observed in many futures markets. In Section 2.3 we describe the seasonal forward curve model which introduces the seasonal premium into futures prices and hence is able to capture seasonalities present in forward curves such as those shown in Figure 2.4.

The dynamic modelling of the forward curve (i.e. the description of its random movements over time) and the understanding of the main fundamental factors behind its evolution have been the subject of extensive research, both by practitioners and academics. A proper model of the dynamics of the forward curve is an essential input in most derivative pricing models, scenario simulation and risk management applications. A successful forward curve model should be able to match current observed forward curves, generate realistic forward curves containing empirically observed features and ideally should be able to extrapolate the forward curve beyond the longest observed maturity. Moreover, it should be easily and quickly calibrated to the market data. However, just as in case of the Treasuries yield curves, commodity forward curve modelling is a challenging task, due to the inherent multidimensionality of the problem and a non-trivial dependence structure across maturities. In contrast to interest rates, features such as seasonality may further complicate the modelling process.

In the next section we review several forward curve modelling approaches, before proceeding in Section 2.3 to the case of seasonal forward curves. Section 2.4 describes the application of the Principal Component Analysis to commodity forward curves. Section 2.5 introduces several forward curve indicators and Section 2.6 concludes.

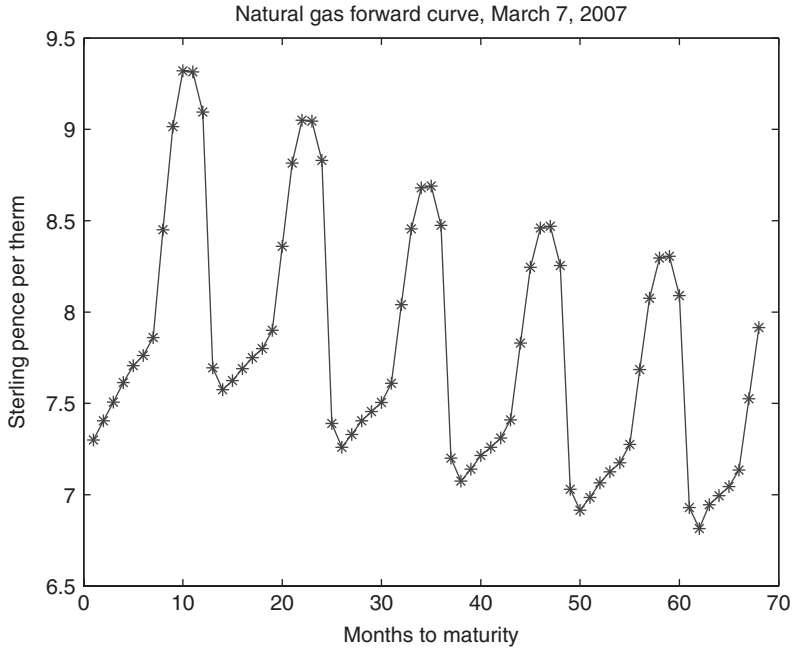


Figure 2.4 UK Natural gas forward curve, 7 March 2007

2.2 FORWARD CURVE MODELS FOR NON-SEASONAL COMMODITIES

Forward curve modelling approaches for commodities can be loosely classified into three main categories. The first one is a martingale-based approach, which models the dynamics of futures prices directly under the risk-adjusted probability measure, making use of the fact that futures prices are martingales under such a measure. One or possibly several risk factors driving the risk-neutral dynamics of the entire forward curve are usually assumed (resulting into so-called one-factor or multifactor models).

Another approach is a static arbitrage-based one, making use of the cash-and-carry arguments and describing the no-arbitrage relationship between the spot and futures prices. In the remainder of this section we shall review these modelling approaches in more detail.

Multifactor models such as described in Clewlow and Strickland (2000) are similar to the celebrated HJM approach for modelling the yield curve (Heath *et al.* (1992)), in that they directly model futures prices under the risk-neutral probability measure.¹ So the following representation is often considered:

$$\frac{dF(t, T)}{F(t, T)} = \sum_{i=1}^n \sigma_i(t, T) dW_i(t), \quad (2.2)$$

¹ As a futures contract does not require an initial investment, under Q its price remains constant on average and there is no drift term representing the average growth on the investment.

where n is the number of risk factors, $W_i(t)$, $i = 1, \dots, n$ are Brownian motions representing sources of uncertainty and $\sigma_i(t, T)$ are volatilities associated with the risk factors (possibly varying deterministically in time and maturity). The Brownian motions can be correlated, but if the risk factors are derived using the Principal Component Analysis (described in Section 2.4), then they are assumed uncorrelated. This approach focusses on the martingale property of $(F(t, T))_{t \leq T}$ under Q , which is useful for derivatives pricing but not for trading strategies involving spot and forward positions, the most common ones in a trading environment, as it says nothing about the actual evolution of the forward curves and hence cannot be calibrated to historical futures market data. We will mainly concentrate in this chapter on the evolution of a forward curve under the statistical probability measure, as we are mainly interested in trading strategies in forward contracts.

In the first approach, the main fundamental factors (sources of uncertainty) are rather arbitrary (e.g., derived by the PCA) and do not always have a clear meaning. Alternatively, more interpretable fundamental stochastic factors can be taken. For example, a popular choice is to take the short-term and long-term price components (e.g. Lucia and Schwartz (2002), Sorensen (2002)), which together determine the commodity's spot price:

$$S(t) = s(t) + X(t) + L(t), \quad (2.3)$$

where $X(t)$ and $L(t)$ are respectively the short- and long-term price components and $s(t)$ is a possible (deterministic) seasonal component of the spot price. Then the futures prices are obtained using the relationship

$$F(t, T) = E_Q(S(T) | \mathcal{F}_t), \quad (2.4)$$

where Q is the risk-adjusted probability measure and $\mathcal{F}_t$ is the information available up to date t . To evaluate the expression (2.4), the dynamics of the fundamental factors under the risk-adjusted probability measure Q should be specified. Usually such dynamics include the market prices of risk (the spot price risk and convenience yield risk), which make these models rather hard to calibrate, as market prices of risk are not observable. Moreover, the forward curves derived from (2.4) using this approach usually do not match observed forward curves, which is obviously a problem.

The static arbitrage approach is based on the no-arbitrage assumption and implies that the futures price of a commodity (or any other asset) must be equal to the cost of acquiring the physical commodity and carrying it until the future's maturity. For financial assets, this means that the futures price must exceed the spot price by the interest accumulated on the investment until the maturity date (minus eventual dividends, in case of stocks). Physical commodities, unlike financial assets, cannot be stored without cost; hence, for them, the cost of carry is also determined by the accumulated storage costs. However, these arguments would imply that commodity forward curves should always be in contango, i.e., futures prices for longer maturities should always be higher than those for shorter maturities and the spot price. In practice, this is rarely the case; in fact, the crude oil futures market historically has been in backwardation approximately 75 % of the time. To cope with this observation, the notion of *convenience yield* was introduced by Kaldor (1939) and Working (1948). It is defined as the premium received by the owner of the physical commodity and

not accruing to the holder of a futures contract written on it. This concept leads to the famous cost-of-carry relationship:

$$F(t, T) = S(t)e^{[r(t)+c(t)-\tilde{y}(t)](T-t)}, \quad (2.5)$$

where $r(t)$ is the riskless interest rate on the date t , $c(t)$ is the storage cost (per unit of time and per dollar worth of commodity) and $\tilde{y}(t)$ the convenience yield, all continuously compound. Often, the convenience yield is defined net of storage costs: $y(t) = \tilde{y}(t) - c(t)$, in which case the cost-of-carry relationship becomes

$$F(t, T) = S(t)e^{[r(t)-y(t)](T-t)}. \quad (2.6)$$

In its early versions, the cost-of-carry relationship (2.6) included a constant or deterministic (but time-varying) convenience yield, such as in Brennan and Schwartz (1985). The next step was to define the convenience yield as a function of the spot price. More realistic versions of (2.6) (e.g. those in Gibson and Schwartz (1990), Litzenberger and Rabinowitz (1995)) included the convenience yield as a stochastic process. Some authors stressed a time-spread optionality embedded in the convenience yield (discussed in e.g. Routledge *et al.* (2000)). To emphasize this option-like feature of the convenience yield, it should be considered as a function of maturity T (or time to maturity $T - t$), as well as time t : $y = y(t, T)$, as suggested in Borovkova and Geman (2006). This representation is particularly useful in the case of seasonal commodities, as we show in the next section.

Obtaining a dynamic model for the forward prices from the cost-of-carry relationship (2.6) goes as follows. The spot price $S(t)$ and the convenience yield $y(t)$ are considered as the fundamental stochastic factors driving the forward curve's evolution. A stochastic model is assumed for the joint dynamics of $S(t)$ and $y(t)$ and then the corresponding dynamics for the futures prices is obtained by substituting the expressions for $S(t)$ and $y(t)$ into the cost-of-carry relationship (2.6). Sometimes the interest rate can be added as another fundamental factor, as in Milrersen and Schwartz (1998). The cost-of-carry relationship (2.6) holds under any probability measure, so for the purposes of derivatives pricing, the risk neutral dynamics of futures prices can be obtained from the risk-neutral dynamics of the fundamental factors.

None of the approaches described above is directly suited to modelling seasonal forward curves, often observed in energy and agricultural markets. One attempt has been made by Sorensen (2002) and Lucia and Schwartz (2002), who consider a seasonal component of the spot price that subsequently enters the expression for the forward prices in a rather opaque way. However, seasonality in forward prices obtained in this way does not match the empirically observed seasonal patterns present in the forward curve, especially the fact that the forward price's seasonality is a function of the contract maturity T and not so much of the current date t . In the next section we introduce an explicit modelling of maturity-related seasonalities in commodity forward curves. Ours is a two (or three) factor model, where the spot price is replaced by a more stable quantity and the convenience yield is considered net of deterministic seasonal effects.

2.3 THE SEASONAL FORWARD CURVE MODEL AND ITS EXTENSIONS

Forward prices of seasonal energy commodities such as natural gas and electricity are influenced by seasonal demand due to heating or airconditioning; agricultural commodities are influenced by seasonal supply (harvest). These constraints result into positive seasonal premia attached to calendar months of high demand or low supply. Furthermore, the commodity spot price can be unreliable or unavailable, so it is not always a good candidate to be the main fundamental factor driving the forward curve. In this section we explore these observations within the framework of the “seasonal forward curve model”. In Section 2.3.1 we describe the original two-factor model proposed by Borovkova and Geman (2006) and apply it to seasonal energy commodities such as natural gas and electricity. In Section 2.3.2 we introduce an important generalization of the original model. In the generalized seasonal forward curve model, we introduce three stochastic factors, in order to allow for a trend in a seasonal forward curve.

2.3.1 Seasonal cost-of-carry model

For clarity of exposition, we will assume that the forward curve contains liquid maturities up to one year (12 months) or an integer number of years. We define the first fundamental factor as *the geometric average of the observed forward prices at date t* :

$$\bar{F}(t) = \sqrt[N]{\prod_{T=1}^N F(t, T)}, \quad (2.7)$$

where N is the most distant maturity and $N = 12 \times k$ ($k = 1, 2, \dots$). This assumption assures that $\bar{F}(t)$ is a non-seasonal quantity, which is another advantage of using it instead of the (possibly seasonal) spot price.

Next, we define the seasonal premia $s(M)$, $M = 1, \dots, 12$, attached to calendar months, as the collection of 12 long-term average premia with respect to the average forward price $\bar{F}(t)$. We assume that the vector $(s(1), \dots, s(12))$ is deterministic and require that $\sum_{M=1}^{12} s(M) = 0$.

Finally, we write the *seasonal cost-of-carry model* describing the relationship between the prevailing forward price $\bar{F}(t)$ and the futures price $F(t, T)$:

$$F(t, T) = \bar{F}(t) e^{[s(T) - \gamma(t, T-t)(T-t)]}, \quad (2.8)$$

where the quantity $\gamma(t, T - t)$, defined by the relationship above, is called the *stochastic premium* at date t (as opposed to the deterministic seasonal premium $s(T)$), associated with the maturity T . We denote the *time to maturity* $\tau = T - t$ and write $\gamma(t, \tau) = \gamma(t, T - t)$, to emphasize that the stochastic premium γ is indexed by the time to maturity τ rather than the maturity date T (as is the case for the deterministic seasonal premium $s(T)$).

We define $s(T)$ for maturity T as follows: $s(T) = s(M)$ if the maturity T is in the calendar month M ($M = 1, \dots, 12$). Note that, in this way, futures maturing in, for

example February 2009 and February 2010, get the same seasonal premium $s(2)$ associated with the month of February. Furthermore, note that we define $s(M)$ as an absolute quantity (expressed in percentage) and not as a rate. This reflects that the seasonal premium does not accumulate with distant maturities, but is only associated with the calendar maturity month. In this way, the seasonal premium for the February contract is the same, be it the front month contract or the one maturing in a year's time.

To summarize, futures maturing in a particular calendar month can be either at a premium or a discount relative to the average level of the forward curve. This is a periodic, i.e., seasonal effect, which is associated with periods of high demand or low supply. This premium is summarized in the non-stochastic quantity $s(M)$, which can be consistently estimated from the historical data.

The quantity $\gamma(t, \tau)$, on the other hand, is a stochastic process in t , indexed by the time to maturity τ , and summarizes all deviations of the actual observed futures prices from the anticipated futures prices given by $\bar{F}(t)e^{s(T)}$. Its expected value is zero, since all systematic deviations of futures prices from $\bar{F}(t)$ are embedded in the seasonal premium $s(T)$. So we will model it by the Ornstein-Uhlenbeck process with zero mean. We call $\gamma(t, \tau)$ the *stochastic cost of carry*, or *stochastic premium devoid of seasonal premium*, for time to maturity τ . Factors influencing the function γ are, for example, levels of stocks, political news and events or unusual weather conditions.

The seasonal cost-of-carry model (2.8) is related to the traditional cost-of-carry models such as the one exhibited in Brennan and Schwartz (1985) and many other papers. To our best knowledge, it is the only model that uses the average forward price $\bar{F}(t)$ instead of the spot price $S(t)$ as the first state variable. Accordingly, the traditional convenience yield introduced by Kaldor (1939) in the Theory of Storage is replaced by the stochastic premium $\gamma(t, \tau)$. This stochastic premium can be seen as the “relative convenience yield” (with respect to its average across all maturities), net of the scaled seasonal premium:

$$y(t, T) - \frac{1}{N} \sum_{M=1}^N y(t, M) = \gamma(t, \tau) - \frac{s(T)}{\tau},$$

where $y(t, T)$ is the traditional convenience yield. The stochastic premium γ and the seasonal cost-of-carry model have the merit of being applicable to non-storable commodities such as electricity. For these commodities, the traditional cost-of-carry model and the convenience yield do not make sense, as the absence of storability prevents any cash-and-carry arguments (see Eydeland and Geman (1998)). Moreover, the seasonal cost-of-carry model does not imply any convergence, smooth or not, of the futures price to the spot price, a property that is unobserved in a number of commodity markets.

Another advantage of having the average forward price $\bar{F}(t)$ as the first fundamental factor is that it is a non-seasonal quantity by construction, and hence can be modelled by a stationary stochastic process of the model builder's choice, e.g., mean reverting or geometric Brownian motion, or including jumps. The quantity $\bar{F}$ gives a very good proxy for the overall level of the futures market, while being more visible and less unstable than the spot price. However, it can happen that, in forward curves spanning several years, the average forward price differs for different years ahead, as it reflects the market anticipation of the remaining reserves left at that time for exhaustible commodities such as copper, crude oil or uranium. Below we introduce an important generalization of the model, which deals with such issues.

Note that the convergence property of the futures price to the spot price (2.1) provides the following expression for the spot price within the seasonal cost-of-carry model:

$$S(t) = \bar{F}(t)e^{s(t)}. \quad (2.9)$$

Hence, if the spot price is not available in a particular market, the above relationship can be used to define a valuable proxy for the spot price.

Returning to the dynamics of the state variables, the stochastic premium can be arguably represented by an Ornstein-Uhlenbeck process with zero mean:

$$d\gamma(t, \tau) = -a^{(\tau)}\gamma(t, \tau)dt + \eta^{(\tau)}dW(t), \quad (2.10)$$

where the mean-reversion speed and the volatility parameters may differ per maturity T . In the above specification, the entire term structure of $\gamma(t, \tau)$ is driven by the same Brownian motion $W(t)$; however, more sources of uncertainty can be easily incorporated. The average forward price $\bar{F}(t)$ can be modelled by any appropriate diffusion process, uncorrelated to the process driving the term structure of $\{\gamma(t, \tau)\}_\tau$. For example, a mean-reverting process can be used, which is a popular choice for commodities. However, in recent years most commodities have experienced massive price increases. Geman (2005) argues that, in the case of crude oil prices, “mean-reversion is dead”. This may be true for other commodities such as copper or uranium. In these cases, one can use other model specifications for $\bar{F}(t)$, such as a geometric Brownian motion or a mean-reverting process with a deterministic (or even stochastic) trend.

The seasonal cost-of-carry model (2.8) provides the relationship between the fundamental factors and the futures contracts prices. So the dynamics and the distributions of the futures prices can be obtained from the dynamics of the fundamental factors, by substituting them into (2.8). For example, the above specification (2.10) of $\gamma(t, \tau)$ and a mean reverting diffusion for the log-average forward price $X(t) = \ln \bar{F}(t)$

$$dX(t) = \alpha(m - X(t))dt + \sigma d\bar{W}(t) \quad (2.11)$$

lead to the following futures price dynamics:

$$d(\ln F(t, T)) = [\alpha(m - X(t)) + \gamma(t, \tau)(a^{(\tau)}\tau + 1)]dt + \sigma d\bar{W}(t) - \eta^{(\tau)}\tau dW(t), [\tau = T - t.] \quad (2.12)$$

So in this case, the futures price $F(t, T)$ is lognormally distributed, with mean $\bar{F}(t)e^{s(t)}$ and the volatility term structure defined by

$$\delta(t, \tau) = \sqrt{\sigma^2 + (\eta^{(\tau)}\tau)^2}. \quad (2.13)$$

This is the description of the futures prices under the statistical probability measure P . For derivatives pricing, only the risk neutral dynamics are relevant. These are obtained by specifying the risk neutral dynamics of the fundamental factors and using again the equality (2.8). Now, under the risk neutral measure Q , futures prices are martingales, i.e., their drift is identically zero. This results into a restriction on the model parameters under Q , similar to the HJM condition (Heath *et al.* (1992)) on Treasury forward rates.

The seasonal premium and other model parameters can be estimated from the set of historical futures prices. The average forward price can be computed on any day t by formula (2.7) (as said before, it is essential that the futures prices up to one year or an integer number of years are included, in order to eliminate seasonal features). The seasonal premium associated with the month M is estimated according to its definition by

$$\hat{s}(M) = \frac{1}{n} \sum_{t=1}^n [\ln F(t, M) - \ln \bar{F}(t)], \tag{2.14}$$

where n is the number of observations in the historical dataset and we include all futures expiring in the calendar month M (possibly for different years). The series of daily stochastic premia $(\gamma(t, \tau))_{t=1, \dots, n}$ for all available times to maturity τ is computed as

$$\hat{\gamma}(t, \tau) = \frac{\ln(\bar{F}(t)/F(t, T)) + \hat{s}(T)}{\tau}. \tag{2.15}$$

From the obtained series $(\bar{F}(t))_t$ and $(\hat{\gamma}(t, \tau))_t$, the parameters of the associated stochastic diffusion models can be estimated by, for instance the method of maximum likelihood.

Figure 2.5 shows the estimated seasonal premia for UK natural gas, electricity and heating oil futures. Our estimates are based on historical datasets of ICE futures prices from 2001 to 2004.

As expected, futures expiring in winter are at a premium with respect to the average price level, and summer futures at a discount. December gas futures are on average at a 28 % premium, December electricity futures are at 15 % premium. The seasonal premium for heating oil is generally smaller, and is at most 3 %. This reflects a wider availability of storage and transportation for heating oil. Note that similar products traded on the NYMEX may exhibit different seasonal premia, as those reflect the risk aversion of producers and speculators in each particular market to different types of events.

Figure 2.6 shows the volatility term structure of the stochastic premium $\gamma(t, \tau)$, for natural gas and gasoil. Note that it mimics the behavior of the futures prices volatility, which decreases for more distant maturities (the so-called Samuelson effect).

Figure 2.7 shows the realization of the natural gas and gasoil stochastic premium $\gamma(t, \tau)$, for $\tau = 2$, i.e. for two months to maturity. The mean of γ is indeed zero and the graph resembles a mean-reverting process.

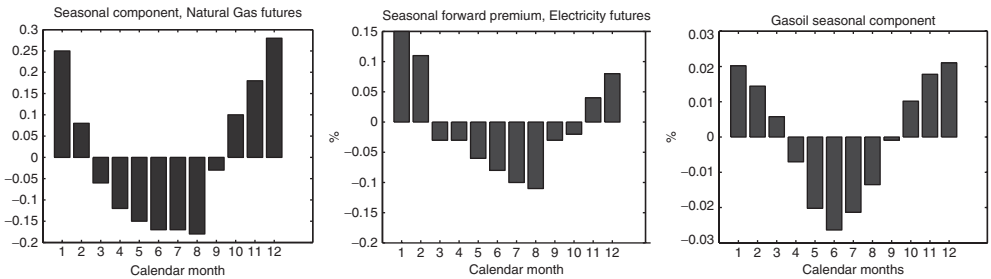


Figure 2.5 Natural gas, electricity and heating oil seasonal premia

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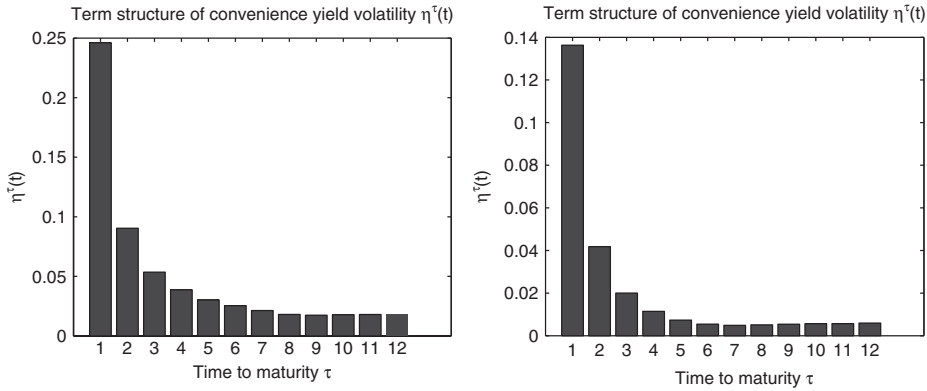


Figure 2.6 Volatilities of natural gas and gasoil stochastic convenience yields

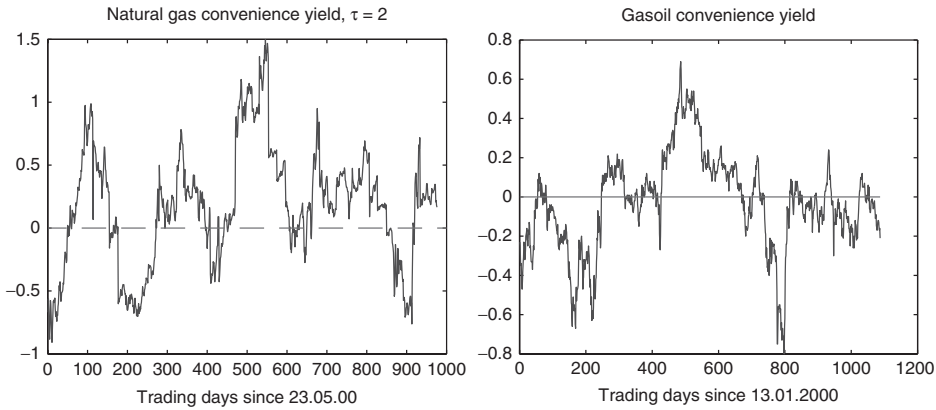


Figure 2.7 Natural gas and gasoil stochastic convenience yields, $\tau = 2$

Applications of the seasonal cost-of-carry model include the generation of realistic seasonal forward curves for risk management and scenario simulations. Moreover, this model can be used for extending the observed forward curves beyond the most distant liquid maturity, a necessary task in the valuation of physical assets such as gas storage facilities or refineries. The generalization of the model, introduced in the next paragraph, is particularly useful for this purpose. The formulation of the fundamental factors' risk neutral dynamics provides a unifying framework for pricing derivatives that depend on the entire forward curve, such as calendar spread options.

2.3.2 The generalized seasonal forward curve model

Recall that, above, we defined $\bar{F}(t)$ as the geometric average of all available forward prices or, in other words, $\ln \bar{F}(t)$ as the arithmetic average of all log-futures prices:

$$\ln \bar{F}(t) = \frac{1}{N} \sum_{T=1}^N \ln F(t, T), \quad (2.16)$$

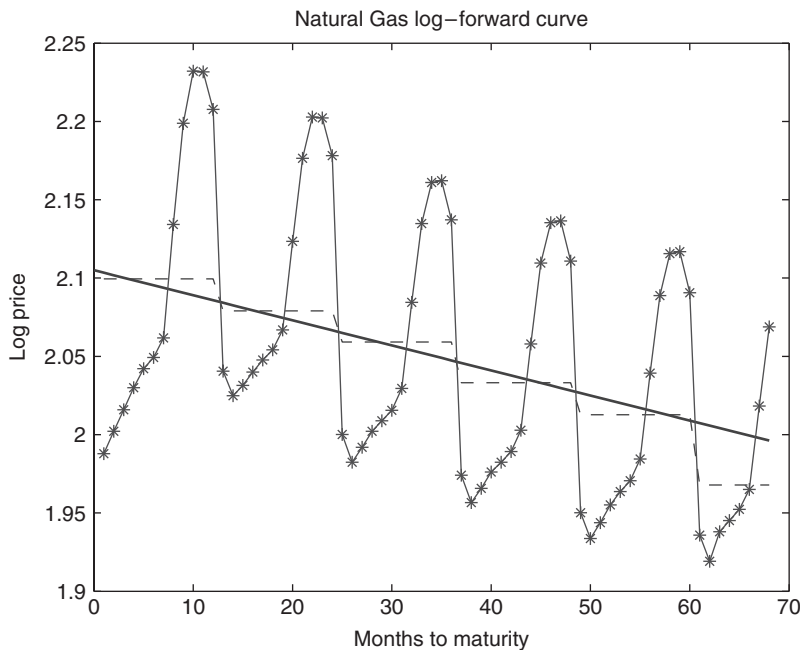


Figure 2.8 NYMEX Natural Gas forward curve, three factor model

where N is the longest traded maturity. In cases such as depicted in Figure 2.4, we can define the (geometric) average forward prices $\bar{F}_i(t)$ for each year i present in the forward curve as

$$\ln \bar{F}_i(t) = \frac{1}{12} \sum_{T=12(i-1)+1}^{12i} \ln F(t, T). \tag{2.17}$$

To obtain a multiplicative model similar to (2.8), we first go from forward prices to their logarithms, so we consider the log-forward curve $(\ln F(t, T))_T$. Figure 2.8 shows the logarithm of a forward curve of NYMEX Natural Gas futures, together with a collection of average yearly forward log-prices $(\ln \bar{F}_i(t))_i$. These average log-prices are highly correlated and move for a large part together, so defining them as separate fundamental stochastic factors does not really make sense, not to mention the increase in the problem’s dimensionality. Instead, we suggest extending the number of fundamental factors by just one extra factor, in the following way.

Note that the collection $(\ln \bar{F}_i(t))_i$ defines a step function (in terms of maturities) approximating the logarithm of the forward curve. A more general way to approximate the overall shape of the log-forward curve on day t is to fit a least-squares straight line through all available forward log-prices:

$$X_T(t) = X_0(t) + \alpha(t)T, \tag{2.18}$$

where the parameters $\alpha(t)$ (slope) and $X_0(t)$ (intercept) are estimated on each day t by the ordinary least squares method. This straight line approximation is illustrated in Figure 2.8.

Note that equation (2.18) describes, at date t fixed, $X_T(t)$ as an affine function of the maturity T .

Denote $\bar{F}_0(t) = \ln X_0(t)$ and $\bar{F}_T(t) = \ln X_T(t)$. The relationship (2.18) implies that the overall shape of the forward curve is approximated by an exponential function given by

$$\bar{F}_T(t) = \bar{F}_0(t)e^{\alpha(t)T}, \quad (2.19)$$

where $\alpha(t)$ evolves randomly over time.

If the slope $\alpha(t)$ on day t is negative, then the forward curve is in overall backwardation, such as in Figure 2.4, where the seasonal premium is still clearly present. The positive slope parameter $\alpha(t)$ defines the contango forward curve shape. If the slope parameter is not significantly different from zero, then we are in the situation of the initial seasonal cost-of-carry model of the previous section, with the parameter $\bar{F}_0(t)$ being exactly the geometric average of the forward curve $\bar{F}(t)$.

The seasonal deterministic premia $s(M)$ ($M = 1, \dots, 12$) are now defined as *the average premia, expressed in %, in futures expiring in the calendar month M with respect to $\bar{F}_T(t)$ described in (2.19) and defining on day t the overall shape of the forward curve.*

The new formulation of the generalized seasonal cost-of-carry model is then

$$F(t, T) = \bar{F}_0(t)e^{\alpha(t)T + s(T) - \gamma(t, T-t)(T-t)}, \quad (2.20)$$

where $\gamma(t, \tau)$ is again the stochastic forward premium, defined by the relationship (2.20) above.

The slope $\alpha(t)$ and $\bar{F}_0(t)$ (or its logarithm $X_0(t)$, which is the intercept), defining together the straight line (2.18), are now the two new fundamental factors of our generalized seasonal forward curve model. The third one remains the stochastic premium $\gamma(t, \tau)$, defined by the relationship (2.20).

The two new fundamental factors $\bar{F}_0(t)$ and $\alpha(t)$ have a well-known interpretation, closely related to the Principal Component Analysis of the forward curve, described in Section 2.4. These factors reflect the *level and slope* of the forward curve on date t and contain most of the forward curve's variability. The remaining stochastic variation of the forward curve, in particular its curvature, is now summarized in the stochastic forward premia $\gamma(t, \tau)$. We postulate that all three fundamental factors are pairwise uncorrelated. Both $\bar{F}_0(t)$ and $\alpha(t)$ are uncorrelated to $\gamma(t, \tau)$ because the latter quantity accounts for unexpected shocks in the commodity market, such as extreme weather conditions or disruption in supply. Furthermore, we can view $\bar{F}_0(t)$ as being uncorrelated to $\alpha(t)$ (the slope of the affine function of maturity). The theory of normal backwardation ($\alpha(t)$ negative when $\bar{F}_0(t)$ high) appears to be essentially invalidated by the current commodity markets, as $\bar{F}_0(t)$ represents the current position held by many market participants, including new entrants, and the slope $\alpha(t)$ carries long-term views of the fundamental players.

The dynamics of the new fundamental factor $\bar{F}_0(t)$ can be specified by a model builder according to the observed empirical features deduced from historical data. For example, it can be a mean reverting process with a constant or increasing level of mean-reversion, or a Geometric Brownian motion. The slope $\alpha(t)$ and the stochastic premium $\gamma(t, \tau)$ should be modelled as Ornstein-Uhlenbeck processes mean reverting to zero. Since we continue to observe, including in the recent past, forward curves that change from backwardation to contango and vice versa, we believe that the choice for $\alpha(t)$ of Ornstein-Uhlenbeck process mean-reverting to zero is a reasonable one.

The fundamental factors' dynamics specifications lead to the dynamics of forward prices via the generalized seasonal cost-of-carry relationship (2.20). Portfolio strategies involving spot and forward positions will be devised using the representation under the real probability measure P . For the purposes of derivatives pricing, all dynamics should be specified under the risk-adjusted probability measure Q . Equation (2.20) can be viewed as written under the measure Q , hence any risk-neutral dynamics for the three fundamental factors imply the corresponding dynamics and distributions for the futures prices.² With these in hand, derivatives such as futures options or calendar spread options can be easily valued by a Black-Scholes-like approach or by Monte Carlo simulations.

2.4 PRINCIPAL COMPONENT ANALYSIS OF A FORWARD CURVE

Forward curve modelling is an inherently multivariate problem: there are as many variables (futures prices) as the number of traded futures contracts with different maturities (e.g., 72 for NYMEX Natural Gas futures). In applications such as risk management, considering them as separate (but correlated) risk factors is rather excessive since these futures prices move, for a large part, together. The well-known and powerful statistical multivariate technique of Principal Component Analysis allows us to concentrate only on a few main uncorrelated *linear combinations* (i.e. principal components) of the forward prices (which may represent the main risk factors), which together describe virtually all possible forward curves or their evolutions.

In studies involving a large number of observed variables, it is often convenient to consider a smaller number of linear combinations of these variables, which summarize their main features. *Principal components* are such uncorrelated linear combinations. There are as many principal components as original variables and, taken together, they explain all the variability in the original data. However, because of the way they are calculated, it is usually possible to consider only a few of the principal components, which together explain *most* of the original variation. The principal components are ordered according to their contribution to the variability of the data: the first component contributes more to the variability than the second one and so on. To visualize the principal components, the coefficients of these linear combinations (i.e. of the principal components) are used. They are called the *principal component loadings* and provide a convenient summary of the influence of the original variables on the principal components (or, equivalently, the response of each original variable to a change in the corresponding principal component), and thus a useful basis for interpretation.

A forward curve of practically any shape can be constructed by combining three simple shapes: the so-called *level, slope and curvature*. Consequently, the dynamic of the forward curve is mainly determined by the evolution of these three main factors: parallel shift of the entire forward curve, "tilting" of the forward curve and changes in its curvature. This was first established by Litterman and Scheinkman (1991) for the US government bond yields and subsequently applied to commodities by Cortazar and Schwartz (1994), who studied the evolution of copper forward curves. It turns out that, for non-seasonal commodities such as crude oil, the first three principal components of the historical futures price returns

² However, as said before, the martingale property of futures prices under the risk-adjusted probability measure leads to certain restrictions on drifts and volatilities of the fundamental factors under Q .

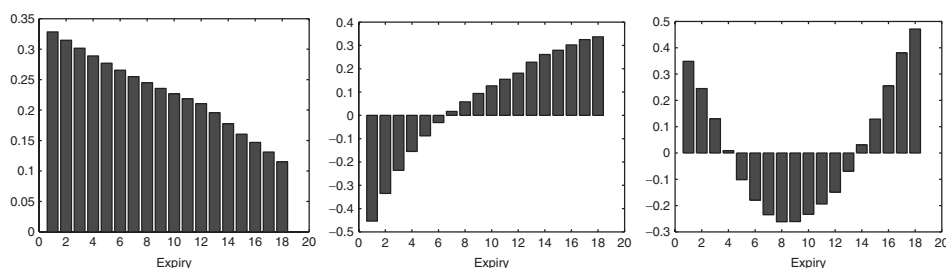


Figure 2.9 Loadings of the first three principal components, oil futures prices

correspond exactly to these three main factors: the level, slope and the curvature. This is shown in Figure 2.9, which depicts the loadings of the first three principal components for the historical returns of Brent oil futures prices for the first 18 maturities.

These first three principal components explain approximately 95 % of the original variation in the forward curves. Hence, they provide an excellent way of extracting most of the essential information and expressing it via a small number of factors. The economic interpretation of these factors is rather straightforward: for energy futures (and, to a lesser extent, for other commodities) a parallel shift of the forward curve is caused mainly by changes in global economy, political situation, and exploration techniques. A slope change is caused by changes in expected long-term price or by a change in the “convenience premium”, or convenience yield, discussed above, on holding a physical commodity. The curvature of the forward curve is related to volatilities of futures prices, as well as long-term vs. short-term expectations on supply and demand.

Depending on a given application, the PCA can be performed either for historical futures prices or returns. For simulations or for generating trading indicators (discussed in Section 2.5), PCA is applied directly to the futures prices, as is done, for example, in Weron (2006) for simulating electricity forward prices. For risk management and derivatives pricing applications, the main risk factors and corresponding volatilities can be conveniently summarized by the principal components of the historical futures price returns. This significantly reduces the number of risk factors in futures portfolios and simplifies Value-at-Risk calculations. For applications of principal components in risk management for bond portfolios see e.g., Golub and Tilman (1997), Singh (1997), and Borovkova (2006) for commodity portfolios. In multifactor forward curve models (such as that of Clewlow and Strickland (2000), represented by the equation (2.2)), the first three principal components of the forward curve are taken to be the three fundamental factors, and corresponding volatility functions $\sigma_i(t, T)$ (suitably parameterized) are estimated from the historical data.

For seasonal commodities, the application and interpretation of the principal component analysis is not straightforward. PCA cannot be applied to historical forward curves directly (although many authors have attempted to do so): the resulting principal components will reflect the harmonics at the main frequencies present in the periodical pattern. As a result, many principal components might seem important in explaining the forward curve’s variability (and not just the first three): for example, Koekebakker and Ollmar (2001) report that as many as 10 principal components are needed to adequately explain the electricity forward curve’s variability, which is obviously a too high-dimensional problem. We would like to stress that this is caused by maturity-related seasonalities found in natural gas forward prices and volatilities.

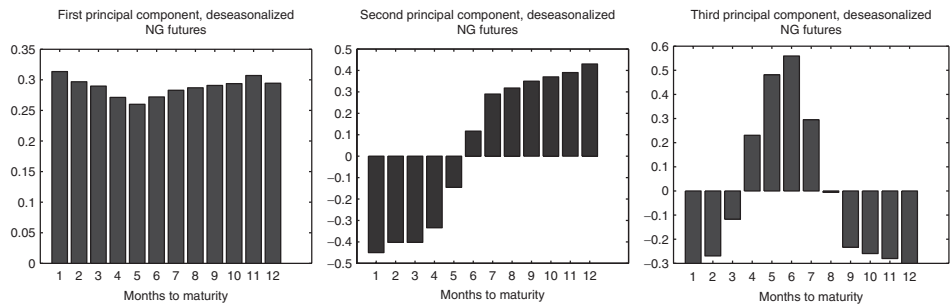


Figure 2.10 Loadings of the first three principal components, deseasonalized natural gas futures prices

One way to deal with prominent seasonalities when applying PCA to seasonal forward curves is to apply PCA separately to historical futures prices maturing in different seasons (an approach adopted in e.g., Tolmasky and Hindanov (2002)), or even separately for each maturity month (as in Blanco *et al.* (2002)). However, this results in over-parametrization. Alternatively, the historical forward prices and returns can be deseasonalized before applying the principal component analysis. For applications such as forward curve simulations, the seasonal premia ($s(M)$, $M = 1, \dots, 12$) can be estimated in the way described above and the PCA applied to the deseasonalized historical forward prices $\tilde{F}(t, T) = F(t, T)e^{-\hat{s}(T)}$, where $\hat{s}(T)$ is the estimated seasonal premium corresponding to the maturity month T .

We applied PCA to the deseasonalized historical natural gas forward curves. The factors analogous to the level, slope, and curvature are clearly present, as Figure 2.10 shows.

For risk management and derivatives pricing applications, futures price returns should be also deseasonalized, since futures price volatility can also exhibit seasonal (maturity-related) patterns (for example, natural gas futures prices are more volatile for futures expiring in months of high demand, i.e. winter months in UK). This can be done by first estimating the seasonal component of the futures price volatility σ_M , $M = 1, \dots, 12$ for all calendar maturity months, and then applying the PCA to the deseasonalized returns $\tilde{r}(t, T) = r(t, T)/\hat{\sigma}_T$, where $r(t, T)$ is day- t return on futures with maturity T and $\hat{\sigma}_T$ is the estimated seasonal volatility component corresponding to the maturity month T . Then the main risk factors extracted by the deseasonalized PCA can be subsequently used in portfolio risk analysis and derivatives pricing, in the same way as for non-seasonal commodities.

2.5 FORWARD CURVE INDICATORS

We presented above a detailed analysis of commodity forward curves and discussed the quantification of a forward curve's features by means of seasonal premia and principal components. However, in a trading environment there is often a preference towards expressing information in the form of technical indicators (such as e.g. Relative Strength Index). An indicator quantifies a certain feature of the spot or forward market by means of a single number, extracted from current market prices. Such indicators can be easily monitored on a daily or intra-day basis and used for developing trading strategies. There are many technical indicators developed for asset spot prices which are routinely applied to commodity spot

prices. However, trading indicators obtained on the basis of forward prices are not that common. Here we discuss several such indicators and outline their potential use.

As we noted above, forward markets for non-seasonal commodities such as crude oil are mainly in two states: backwardation or contango. Whether the market is in backwardation or in contango depends on many fundamental factors. The depth of backwardation or contango (characterized by the steepness of the forward curve) or an anticipated switch from one state to the other is important for market participants: it influences their trading and production decisions. Hence, it is important to detect or forecast such a switch as early as possible. Often, the difference between futures prices with two closest maturities (the so-called *first intermonth spread*) is used for measuring the strength of backwardation or contango and for detecting changes in forward markets. Fundamentals of a commodity market are ultimately responsible for any changes in forward curves, so a change can be anticipated by studying the market fundamentals. In practice, however, traders are interested in working with indicators derived from the futures prices alone, such as the first intermonth spread. The first intermonth spread (or any other calendar spread) is rather inefficient as an indicator of an overall forward curve shape, as it uses only futures prices with two maturities, while many liquid futures prices are often available. Here we describe two change indicators that use the entire forward curve, i.e. all available futures prices. The first indicator measures the “strength” of the backwardation or contango market, i.e. the steepness of the forward curve. The second indicator is based on the principal component analysis. The application of these indicators to seasonal forward curves can reveal features normally obscured by dominant seasonal effects.

A forward market can be in backwardation or in contango for a prolonged period of time. A change from one state to the other is gradual and takes several days or even weeks. The change from backwardation to contango and vice versa is characterized by an intermediate stage of an almost flat forward curve. This observation is crucial in constructing the change indicators.

Denote $Y(t, T) = F(t, T + 1) - F(t, T)$, $T = 1, \dots, N - 1$, the day- t intermonth spreads, i.e. the differences in futures prices with consecutive maturities (N being the longest liquid maturity). The first candidate for the change indicator is

$$I_1(t) = \sum_{T=1}^{N-1} w_T Y(t, T), \quad (2.21)$$

where $(w_T)_{T=1, \dots, N-1}$ is a suitably chosen collection of weights. $I_1(t)$ measures the steepness of the forward curve on day t and the following three situations can arise:

1. $I_1(t) > 0 \rightarrow$ upward-sloping forward curve (contango);
2. $I_1(t) < 0 \rightarrow$ downward-sloping forward curve (backwardation);
3. $I_1(t) \approx 0 \rightarrow$ flat forward curve (intermediate market state), indicating a possibility of a change.

Futures prices are volatile and, in turn, so is the indicator I_1 . Hence it can get close to zero solely as a result of a random shock. On the other hand, the transition between market states happens gradually. So only a persistent near-zero value of I_1 indicates a market change. Instead of the “raw” indicator I_1 , therefore, we can use the moving average

smoothed indicator $\tilde{I}_1$:

$$\tilde{I}_1(t) = \frac{1}{M} \sum_{i=0}^{M-1} I_1(t-i),$$

where M is a suitably chosen moving window size.

The purpose of the weights (w_T) is two-fold. First, they allow us to scale the intermonth spreads in the indicator (2.21) by their volatilities, to discount for liquidity-related effects. Second, it is possible to give more weight to futures prices with shorter maturities if we believe that the front end of the forward curve “drives” the whole curve. We suggest to take weights of the form:

$$w_T = \frac{\gamma^T}{\sigma(t, T)}$$

for some $\gamma \in (0, 1)$, where $\sigma(t, T)$ is the day- t volatility of the T th intermonth spread, suitably estimated. The choice of γ for a particular commodity can be data-driven, by a variant of cross-validation on the basis of historical data. Applications to the oil futures prices have shown that values of γ in the range $0.8 \div 0.95$ perform quite well.

Another parameter to be chosen is the moving average window M . Choosing M too large leads to a late signal (oversmoothing), while choosing M too small results in the indicator being too noisy (undersmoothing). In practice, we have to balance these two factors. For example, one can use $\tilde{I}_1(t)$ for different values of M (“fast” and “slow” moving averages) at the same time.

The second indicator is based on the Principal Component Analysis of the forward curve. Recall that the three main movements of a forward curve are the parallel shift, “tilt” and the change in its curvature, represented by the first three principal components of the forward curve. When studying transitions from backwardation to contango and vice versa, we are interested in the movement of the second type, i.e. when the forward prices with short and long maturities move in the opposite directions. So the second principal component, corresponding to the “tilt” in the forward curve, can be used to construct a change indicator.

To get rid of the dominant parallel shift component, we first remove the average day- t forward price $\bar{F}(t)$ and consider $\tilde{F}(t, T) = F(t, T)/\bar{F}(t)$ (all $\tilde{F}(t, T)$ are scaled to have mean one, for any date t). Then the *first* principal component of $(\tilde{F}(t, T))_{T=1, \dots, N}$ corresponds to the slope of the forward curve, as shown in Figure 2.11, which depicts the first two principal components of $(\tilde{F}(t, T))_{T=1, \dots, N}$.

We define the principal component indicator by

$$I_2(t) = \sum_{T=1}^N L_T^{(1)} \tilde{F}(t, T), \quad (2.22)$$

where $L_T^{(1)}, T = 1, \dots, N$ are the loadings of the first principal component of $(\tilde{F}(t, T))_{T=1, \dots, N}$ (shown on the left plot in Figure 2.11). Note that the equation (2.22) defines a *projection* of the scaled forward curve $(\tilde{F}(t, T))_{T=1, \dots, N}$ on the first principal

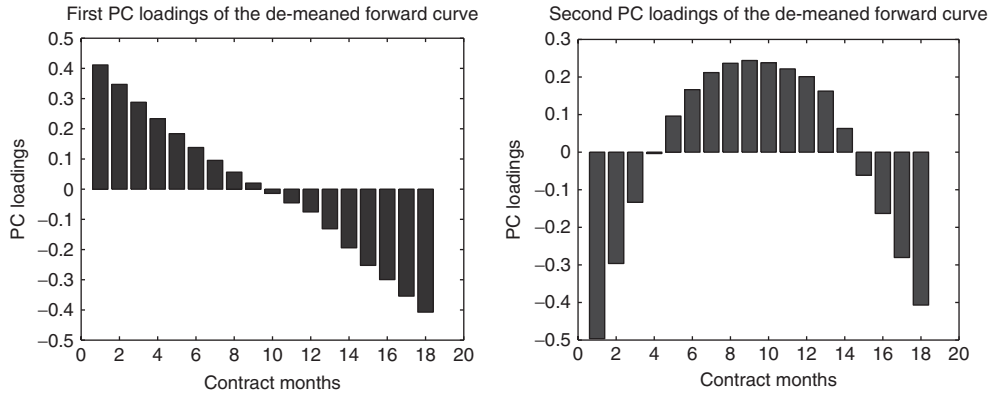


Figure 2.11 First two principal components of $(\tilde{F}(t, T))_T$

component. The corresponding moving average smoothed indicator is then

$$\tilde{I}_2(t) = \frac{1}{M} \sum_{i=0}^{M-1} I_2(t-i),$$

with a suitably chosen moving window length M .

Indicators I_1 and I_2 (and their moving average smoothed versions) can be computed daily and monitored for detecting a switch between backwardation and contango. Loosely speaking, values of indicators close to zero (i.e. in some critical ε -neighborhood of zero) signal a possible switch. A critical value of ε can be specified via the distribution (empirical or theoretical) of the indicators under the null-hypothesis of no change. Alternatively, simulation techniques called the *smooth stationary bootstrap*, introduced in Borovkova (2004), can be used to construct critical regions for the indicators.

For seasonal commodities, the most prominent feature of the forward curve is the maturity-related seasonality, summarized in the seasonal premia $(s(M))_{M=1, \dots, 12}$. The *deseasonalized forward curve* $(F(t, T)e^{-s(T)})_T$ can exhibit features similar to backwardation or contango, or show deviations from the expected seasonal pattern. So for seasonal commodities, we define both indicators I_1 and I_2 in relation to the deseasonalized forward curve $(F(t, T)e^{-s(T)})_T$. More precisely, the weighted strength indicator $I_1(t)$ is defined again by (2.21), with $Y(t, T)$ being the intermonth spreads of the deseasonalized forward curve and with weights $(w(T))_T$ reflecting particular features of a commodity. For instance, the weights for a few shorter maturities can be chosen higher than all others, to reflect most significant deviations occurring in the front end of the forward curve. The principal component indicator is defined as the projection of the deseasonalized (and discounted by $\bar{F}(t)$) forward curve $\tilde{F}(t, T) = F(t, T)e^{-s(T)}/\bar{F}(t)$ on the first principal component of $(\tilde{F}(t, T))_T$.

These indicators can be used in the same way as for non-seasonal commodities, monitoring changes between an overall backwardation or contango state of the forward market, while still allowing for seasonalities. In markets where there is no significant slope in the forward curve (such as the case for shorter forward curves, e.g. for maturities up to 12 months), the

expected deseasonalized forward curve should be approximately flat. Then both indicators (and, to a lesser degree, the projection of the forward curve on the curvature principal component) measure the forward curve’s deviations from the flat shape. This means that values of the indicators far from zero (e.g., outside 95 % confidence bounds around zero) indicate significant deviations of the forward curve from the expected shape given by the dominant seasonal component.

Figure 2.12 shows the Principal Component Indicator for the UK Natural Gas futures prices (01.04.01–01.04.08), together with its overall historical 95 % confidence intervals around zero.

There are several periods when the indicator leaves its 95 % confidence bounds. During these periods, there are significant deviations of the futures prices from the seasonal pattern. For example, around the 1500th observation (March 2007) the indicator values are very high, meaning that during this period a prominent backwardation shape was detected in the forward curve. Such extreme situations are clearly not persistent, so this information can be used for constructing profitable trading strategies.

Tracking the indicators for seasonal commodities can be useful for monitoring futures markets, detecting shifts in fundamentals or possible mispricing of some futures. Trading signals can be generated when the indicators exceed some pre-determined threshold, generated from either historical data or simulations. In the absence of delivery issues, a simple speculative trading strategy during periods of high (and positive) indicator values could be to sell futures with closer expiries, buy futures expiring later, and close all the positions as soon as the indicator returns inside the threshold bounds. The size of the future’s positions can be made proportional to the principal component loadings. The risk exposure of such

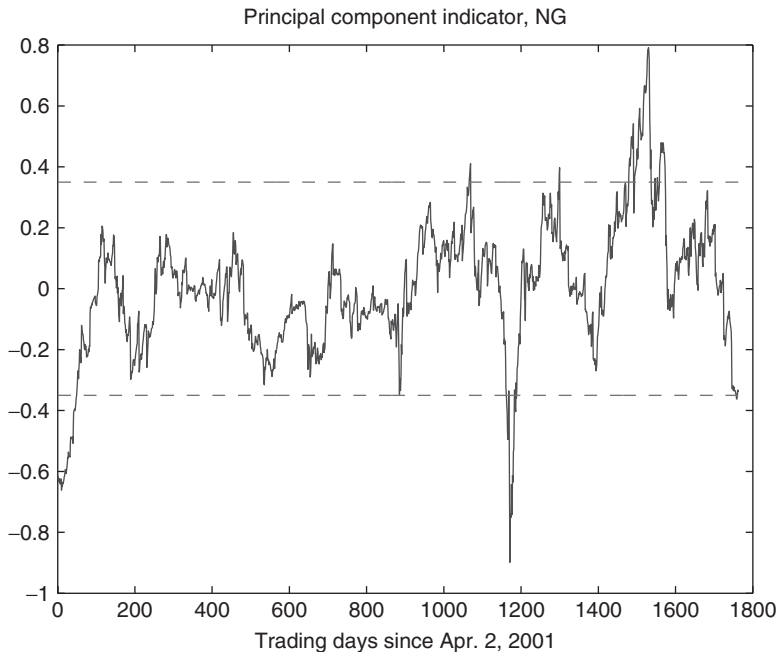


Figure 2.12 Principal Component Indicator for NG futures, 01.04.01–01.04.08

a trading strategy is due to calendar spreads and not to the outright futures prices, which leads to a quite low value-at-risk.

Other (possibly more sophisticated) trading strategies are feasible. For instance, one can also involve the curvature principle component, to actively trade in the middle of the forward curve (middle maturities), in addition to short and long maturities. Additionally, one can exploit autocorrelations present in the daily indicator series: a predictive model (such as an autoregressive one) can be fitted to the historical indicator series and used for forecasting. Generally, the forward curve indicators (together with the estimated seasonal premia) provide valuable information about the state of the futures market and economic fundamentals.

2.6 CONCLUSIONS

Forward curves are of paramount importance in commodity markets. They are essential for financial and physical asset pricing and production planning decisions as they provide information about anticipated price trends. Forward curves obtained from liquid futures markets are essential tools for marking portfolio positions to market and calibrating derivatives pricing models under the risk-adjusted probability measure. Dynamic modelling of forward curves is a challenging task for researchers and practitioners, due to the multi-dimensionality of the problem and particularly complex empirical features of forward curves observed in the markets.

Traditional cost-of-carry arguments are only partially successful in modelling energy forward curves. This is because, on the one hand, the spot price is not always a good image of the current market conditions and, on the other hand, a maturity-independent convenience yield fails to capture seasonal effects often present in energy forward curves. The seasonal forward curve model overcomes both limitations and is capable of capturing prominent characteristic features of seasonal forward curves. The generalized seasonal forward curve model introduced here incorporates the existence of non-seasonal trends in forward prices, while still preserving seasonalities. It is a three-factor model, both parsimonious and sufficiently rich to generate realistic forward curves.

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